

Egységtesztek - Dokumentáció

April 30, 2026

1 unit

- unit namespace
 - Subpackages
 - * unit.dashboard namespace
 - Submodules
 - unit.dashboard.test_auth module
 - unit.dashboard.test_dashboard_db module
 - unit.dashboard.test_management module
 - unit.dashboard.test_orm_dto_convert module
 - * unit.gateway namespace
 - Submodules
 - unit.gateway.test_gateway_db module
 - unit.gateway.test_gateway_utils module
 - unit.gateway.test_key_auth_valid module
 - * unit.shared namespace
 - Submodules
 - unit.shared.test_shared_utils module
 - Submodules
 - unit.conftest module
 - * dashboard_client()
 - * gateway_client()
 - * validated_request()

2 unit.dashboard namespace

2.1 Submodules

2.2 unit.dashboard.test_auth module

2.2.1 test_create_access_token_encodes_payload()

Verify create_access_token generates a JWT token with user data and expiration.

2.2.2 `async test_get_user_from_token_found_user(monkeypatch)`

Verify user extraction returns DTO when token is valid and user exists in database.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace `get_user_by_id` and `convert_orm_to_display_info` helpers.

2.2.3 `async test_get_user_from_token_valid_invalid_missing_user(monkeypatch)`

Verify user extraction from token: valid user found returns DTO, invalid token returns None, user not found returns None.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace `get_user_by_id` and `convert_orm_to_display_info` helpers.

2.2.4 `test_verify_access_token_valid_and_invalid()`

Verify token validation: valid tokens return True, invalid and malformed tokens return False.

2.3 `unit.dashboard.test_dashboard_db` module

2.3.1 `async test_delete_key_archives_and_missing(monkeypatch)`

Verify key deletion archives active key, removes quotas, and raises when key is missing.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to control lookup behavior for keys and quotas.

2.3.2 `async test_delete_user_success_and_guard_rails(monkeypatch)`

Verify user deletion removes related records and blocks missing, administrator, and project-manager deletions.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace admin/project-manager checks and delete helpers.

2.3.3 `async test_is_password_expired_returns_expected_flags()`

Verify password expiry returns true for past timestamp and false for missing or future timestamp.

2.3.4 `async test_permission_role_checks_project_scoped_and_global(monkeypatch)`

Verify member, manager, and administrator checks across scoped, global, and missing-user cases.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace permission and user lookup helpers.

2.3.5 `async test_user_availability_checks_with_and_without_exclusion(monkeypatch)`

Verify email and username availability checks for free, taken, and excluded-owner cases.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace `get_user_by_email` and `get_user_by_username` lookups.

2.4 `unit.dashboard.test_management` module

2.4.1 `async test_enforce_existing_limit_found_and_not_found(monkeypatch)`

Verify limit check passes when exists, raises 404 when not.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace `get_limit_by_id` during the test.

2.4.2 `async test_enforce_existing_quota_target_scenarios(monkeypatch)`

Verify quota target check validates all present entities, raises 404 for missing.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace project/user/key lookup helpers.

2.4.3 `async test_enforce_key_deletion_permission_scenarios(monkeypatch)`

Verify key deletion: admin unrestricted, owner or project manager allowed, others denied.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace admin and ownership lookup helpers.

2.4.4 `async test_enforce_password_change_validity_scenarios(monkeypatch)`

Verify password change: correct current, strength check, no reuse, confirmation match.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace `verify_password` helper during the test.

2.4.5 `async test_enforce_password_strength_valid_and_invalid(monkeypatch)`

Verify strong passwords pass, weak passwords raise with appropriate detail.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture available if password verification helpers need replacing (not used in this specific assertion-based test).

2.4.6 `async test_enforce_quota_creation_permission_scenarios(monkeypatch)`

Verify quota creation: admin for global, project manager for scoped, non-admin denied.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace permission-check helpers (`is_user_administrator`, `is_user_project_manager`).

2.4.7 `async test_enforce_quota_deletion_permission_scenarios(monkeypatch)`

Verify quota deletion: admin unrestricted, project manager for their project, others denied.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace permission and quota lookup helpers.

2.5 `unit.dashboard.test_orm_dto_convert` module

2.5.1 `async test_convert_aggregate_row_to_display_info(monkeypatch)`

Verify aggregate row to DTO conversion maps row values and resolved display names.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace project/user/key lookup helpers used when resolving names.

2.5.2 `test_convert_key_orm_to_display_info_with_overrides()`

Verify key ORM to DTO conversion prefers explicit username and `api_key` overrides.

2.5.3 `test_convert_key_orm_to_display_info_with_related_username()`

Verify key ORM to DTO conversion uses related user username when no override is provided.

2.5.4 `test_convert_limit_orm_to_display_info()`

Verify limit ORM to DTO conversion maps id, name and description.

2.5.5 `test_convert_period_enum_to_display_info()`

Verify period enum to DTO conversion exposes the enum name.

2.5.6 `test_convert_project_orm_to_display_info()`

Verify project ORM to DTO conversion maps identity, name, status and timestamps.

2.5.7 `async test_convert_quota_orm_to_display_info_with_name(monkeypatch)`

Verify quota ORM to DTO conversion maps fields and generated name with lookups.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace lookup helpers (`get_limit_by_id`, `get_user_by_id`, `get_key_by_id`).

2.5.8 `async test_convert_usage_log_orm_to_display_info(monkeypatch)`

Verify usage log ORM to DTO conversion maps values and resolved display names.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace project/user/key lookup helpers used when resolving names.

2.5.9 `async test_convert_user_orm_to_display_info_with_role(monkeypatch)`

Verify user ORM to DTO conversion maps fields and resolved role flags.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace role/permission helper functions (`is_user_administrator`, `is_user_project_manager`, etc.).

3 `unit.gateway` namespace

3.1 Submodules

3.2 `unit.gateway.test_gateway_db` module

3.2.1 `async test_db_key_check_returns_key_and_raises_for_missing()`

Verify API key lookup returns the key record when found and raises `ValueError` when missing.

3.2.2 `async test_db_limit_change_clamps_allocations(monkeypatch)`

Verify quota allocation updates clamp limited quotas and leave unlimited quotas unchanged.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace quota retrieval helpers (e.g. `db_get_quotas_by_key`).

3.2.3 `async test_db_limit_check_and_allocation_handles_common_branches(monkeypatch)`

Verify quota allocation succeeds with no or unlimited quotas and fails when a strict quota would be exceeded.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace quota change and retrieval functions (`db_limit_change`, `db_get_quotas_by_key`).

3.2.4 `async test_get_quota_count_and_get_quotas_by_key(monkeypatch)`

Verify quota counting returns the scalar count and quotas are grouped by their limit IDs.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace module-level helpers such as `_get_limit_ids`.

3.2.5 `async test_refresh_and_expire_quotas_by_batch_update_rows(monkeypatch)`

Verify batch refresh resets quotas and batch expiry marks active quotas as expired.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace time advancing helpers such as `calculate_date_after_period`.

3.3 `unit.gateway.test_gateway_utils` module

3.3.1 `test_calculate_cost_falls_back_to_default_pricing(monkeypatch, capsys)`

Tests that `calculate_cost` falls back to default pricing and logs a warning when `calc_price` raises an exception, and that it returns the default input token price multiplied by the input tokens in usage. Actual calculation uses the `genai_prices` package. As such, tests of `genai_prices` functionality are out of scope, but the fallback should be tested.

- **Parameters:**

- **monkeypatch** (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace pricing dependencies.
- **capsys** (*pytest.CaptureFixture*) – Capture fixture used to inspect fallback warnings.

- **Returns:** None.
- **Return type:** *None*

3.3.2 `test_get_token_count_joins_string_and_content_part_text(monkeypatch)`

Tests that `get_token_count` joins plain string message content and text fields from content-part dictionaries, while ignoring non-text content parts.

- **Parameters:** **monkeypatch** (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace the tokenizer encode function.
- **Returns:** None.
- **Return type:** *None*

3.3.3 `async test_stream_generator_emits_chunks_and_schedules_usage(monkeypatch, validated_request)`

Tests that `stream_generator` emits properly formatted SSE chunks from the raw stream, schedules a usage entry with the correct parameters after iteration, and handles upstream completion without marking failure.

- **Parameters:**
 - **monkeypatch** (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace stream handling dependencies.
 - **validated_request** (*ValidatedRequest*) – Validated request fixture used to drive stream generation.
- **Returns:** None.
- **Return type:** *None*

3.3.4 `async test_stream_generator_emits_error_and_marks_failure(monkeypatch, validated_request)`

Tests that `stream_generator` emits an error chunk and a done chunk when the raw stream raises an exception on the first iteration, schedules a usage entry marked as failed with the correct parameters, and returns early without further iterations.

- **Parameters:**
 - **monkeypatch** (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace stream handling dependencies.
 - **validated_request** (*ValidatedRequest*) – Validated request fixture used to drive stream generation.
- **Returns:** None.
- **Return type:** *None*

3.4 unit.gateway.test_key_auth_valid module

3.4.1 *async* test_check_limits_costs_returns_validated_request_and_blocks_over_quota(monkeypatch)

Verify quota checking returns a validated request when allocation succeeds and raises 429 when allocation fails.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to replace quota module helpers (`get_token_count`, `calculate_cost`, `db_limit_check_and_allocation`).

3.4.2 *async* test_validate_api_key_accepts_valid_and_rejects_invalid_inputs(monkeypatch)

Verify API key validation accepts a valid active key and rejects missing, short, inactive, fingerprint, and signature failures.

- **Parameters:** `monkeypatch` (*pytest.MonkeyPatch*) – Monkeypatch fixture used to override `EXPECTED_KEY_LENGTH`, `db_key_check`, and `verify_key` during the test.

4 unit namespace

4.1 Subpackages

- unit.dashboard namespace
 - Submodules
 - unit.dashboard.test_auth module
 - * `test_create_access_token_encodes_payload()`
 - * `test_get_user_from_token_found_user()`
 - * `test_get_user_from_token_valid_invalid_missing_user()`
 - * `test_verify_access_token_valid_and_invalid()`
 - unit.dashboard.test_dashboard_db module
 - * `test_delete_key_archives_and_missing()`
 - * `test_delete_user_success_and_guard_rails()`
 - * `test_is_password_expired_returns_expected_flags()`
 - * `test_permission_role_checks_project_scoped_and_global()`
 - * `test_user_availability_checks_with_and_without_exclusion()`
 - unit.dashboard.test_management module
 - * `test_enforce_existing_limit_found_and_not_found()`
 - * `test_enforce_existing_quota_target_scenarios()`
 - * `test_enforce_key_deletion_permission_scenarios()`
 - * `test_enforce_password_change_validity_scenarios()`
 - * `test_enforce_password_strength_valid_and_invalid()`
 - * `test_enforce_quota_creation_permission_scenarios()`

- * test_enforce_quota_deletion_permission_scenarios()
 - unit.dashboard.test_orm_dto_convert module
 - * test_convert_aggregate_row_to_display_info()
 - * test_convert_key_orm_to_display_info_with_overrides()
 - * test_convert_key_orm_to_display_info_with_related_username()
 - * test_convert_limit_orm_to_display_info()
 - * test_convert_period_enum_to_display_info()
 - * test_convert_project_orm_to_display_info()
 - * test_convert_quota_orm_to_display_info_with_name()
 - * test_convert_usage_log_orm_to_display_info()
 - * test_convert_user_orm_to_display_info_with_role()
- unit.gateway namespace
 - Submodules
 - unit.gateway.test_gateway_db module
 - * test_db_key_check_returns_key_and_raises_for_missing()
 - * test_db_limit_change_clamps_allocations()
 - * test_db_limit_check_and_allocation_handles_common_branches()
 - * test_get_quota_count_and_get_quotas_by_key()
 - * test_refresh_and_expire_quotas_by_batch_update_rows()
 - unit.gateway.test_gateway_utils module
 - * test_calculate_cost_falls_back_to_default_pricing()
 - * test_get_token_count_joins_string_and_content_part_text()
 - * test_stream_generator_emits_chunks_and_schedules_usage()
 - * test_stream_generator_emits_error_and_marks_failure()
 - unit.gateway.test_key_auth_valid module
 - * test_check_limits_costs_returns_validated_request_and_blocks_over_quota()
 - * test_validate_api_key_accepts_valid_and_rejects_invalid_inputs()
- unit.shared namespace
 - Submodules
 - unit.shared.test_shared_utils module
 - * test_find_project_root_and_missing()
 - * test_hash_and_verify_key()
 - * test_password_hash_and_verify()

4.2 Submodules

4.3 unit.conftest module

4.3.1 dashboard_client()

Return a TestClient instance for unit testing the dashboard.

4.3.2 gateway_client()

Return a TestClient instance for unit testing the gateway.

4.3.3 validated_request()

Return a minimal validated request object for gateway utility tests.

5 unit.shared namespace

5.1 Submodules

5.2 unit.shared.test_shared_utils module

5.2.1 test_find_project_root_and_missing(tmp_path)

Tests that find_project_root can correctly locate the root directory based on the marker and raises FileNotFoundError when the marker is not found.

- **Parameters:** `tmp_path` (*pathlib.Path*) – Temporary filesystem path provided by pytest for creating test directories and markers.

5.2.2 test_hash_and_verify_key()

Test that a key hashed with hash_key can be verified with verify_key, that incorrect keys do not verify and that malformed hashes gracefully return False.

5.2.3 test_password_hash_and_verify()

Tests that a password hashed with hash_password can be verified with verify_password, that incorrect passwords do not verify and that malformed hashes gracefully return False.